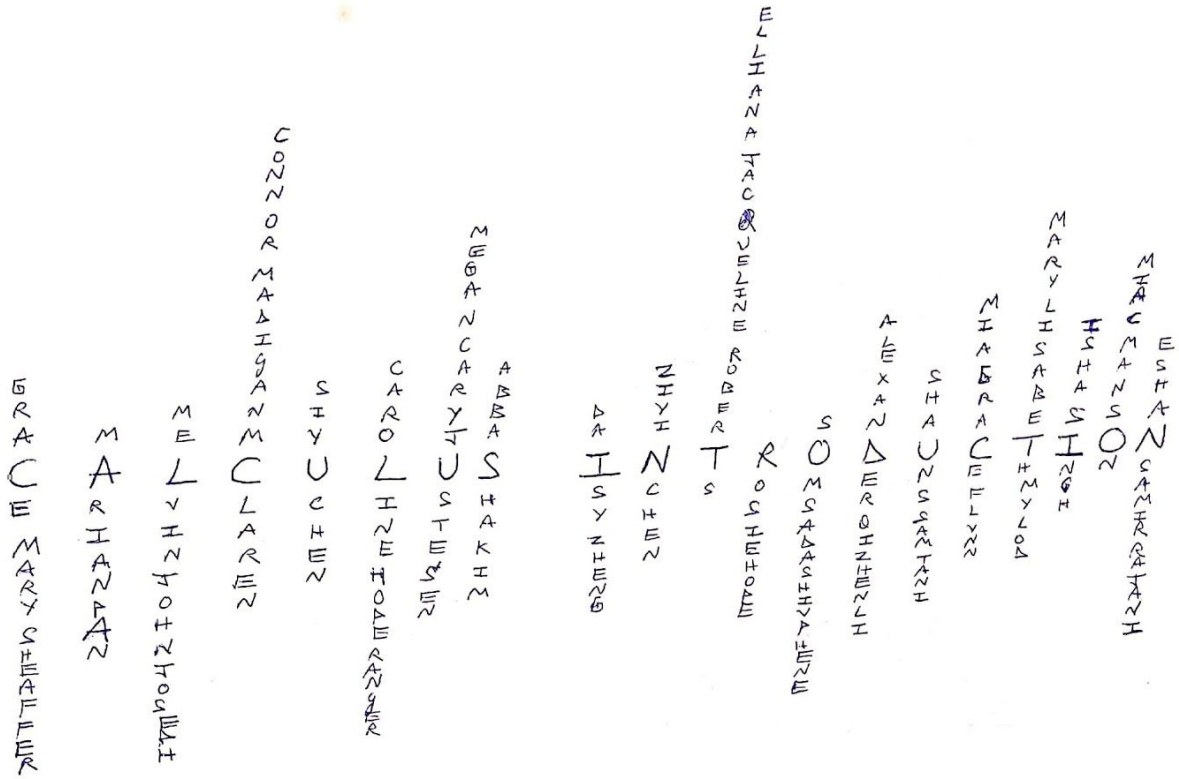
MATH 115 001 WN 2021

1 Introduction



CS Scanned with CamScanner

Introduction: Name-Location-One

- Name: First-Middle-Last
- I-Id-Identity

Communication: 0-Explore-1

Mathematics-as-Child

Calculus: Differential-Function-Integral

Games: Qualifiers-Middle-Final

Practice-Repeat-Practice!

Alright-Alright-Alright

Sets: Point1-Collect-Point2

- { | }
- { x | x }
- { x , y }
- Collection0-Map-Collection1
- Relations: X-(x,y)-Y

Function: Sets-with-Rule

- x-function-f(x)
- x name-function-one value
- n variables-Sum-Mean Value $\times n$
- $\{x_1, x_2, x_3..., x_n\}$- $\{x_1 + x_2 + x_3 + ... + x_n\}$ - Mean $\times n$
- Functions-give-One Meaning
- $\{m, m_2, m_3..., m_n\}$- $\{m_1R + m_2R_2 + ... + m_nR_n\}$ - Centre of Mass $\times Mean\ of\ mass \times n$
- Vertices of n-polygon-Radial Lines- Centre of Polygon
- Domain-Map-Range
- Root-Line-Intercept

Function is I. Function is Introduction. Introduction: Location-Name- One. Domain - each choose one - codmain.

Text. **Function**: rule to give one number for each input number.

Domain: set of all input numbers. Range: set of all output numbers.

If the domain of a function is not specified, we usually take it to be the largest possible set of real numbers. Infinity is not considered to be a real number.

Example: we usually think of the domain of the function $f(x) = x^2$ as all real numbers. However, the domain of the function $f(x) = 1/x$ is all real numbers except zero, since we cannot divide by zero.

Non-Example: $x^2 + y^2 = 1$ is not a function since there is more than one output number(y) for most input numbers(x).

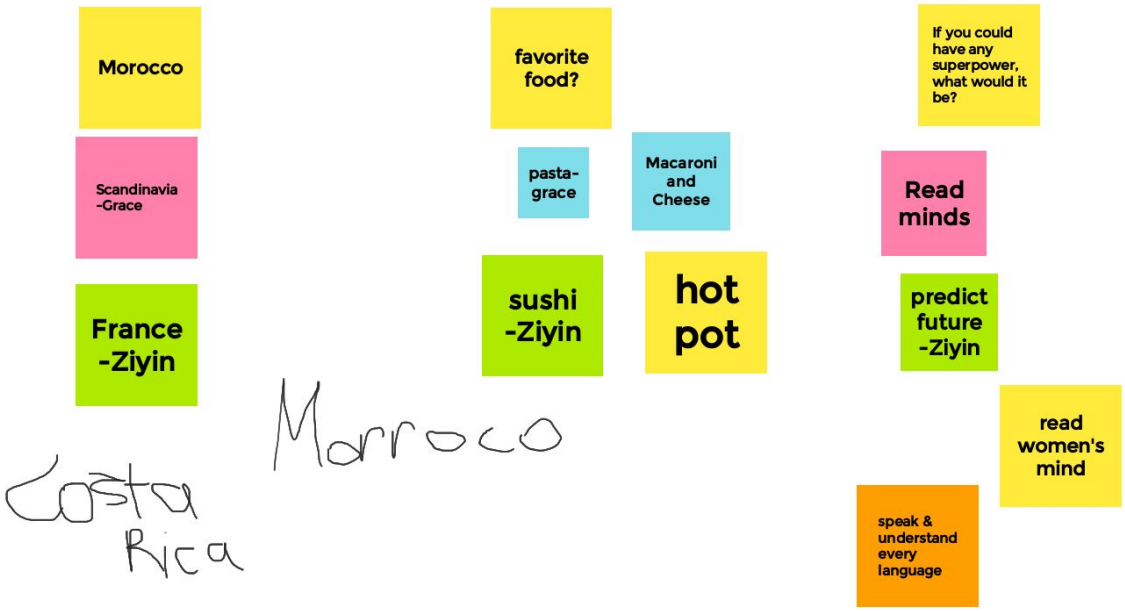
Rule of Four: Read in between the lines. It is just one thing! Multiple Identities representing one thing, communicate with each other.

(Name) Formula- (location Geo)Graphical - (NumericalValue)Table

All of which is Verbal!

Text. Rule of four: table, graph, formula, verbal

[Ice Breaker](), [Ice Breaker.pdf]()



Favorite book/tv show/movie?

Breaking Bad

outmatched

Interstellar

Superbad

star wars

Recommend a song to your group.

after hours

if the world was ending

Gravity- John Mayer

kilby girl

If you could live in any fictional world, which one would you live in?

hogwarts

Star Wars

Marvel universe

Marvel universe

Favorite book/tv show/movie?

Jane the Virgin

Sherlock

Greys Anatomy

vampire diaries

criminal minds

Recommend a song to your group.

grenade by Bruno Mars

lost by Frank Ocean

Feeling Whitney by Post Malone

"Japanese Denim" by Daniel Caesar

Someone to you -BANNERS

If you could live in any fictional world, which one would you live in?

harry potter

avatar (blue people)

star wars

Narnia

[Jamboard 1.1](), [A Jamboard 1.1](), [1.1 Functions.pdf]()

- function

1. Which of the following mathematical objects represent functions?

Function: ouptut one number for each input number

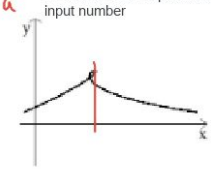
I the rule that sends every non-zero number to its square and sends 0 to 1

II the rule that sends every number to 42

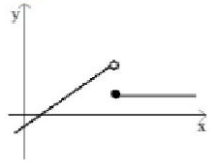
III the rule that, given the weight of an envelope, decides the least number of stamps needed

IV the rule that, given the number of stamps needed, decides the weight of the envelope

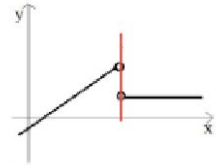
a more than one output number for an input number



V



b No (less than one) output number for an input number



c more than one output number for an input number (x=1 has output y=5 or y=2)

x	y
1	5
1	2
2	3
3	4

VI

x	y
1	5
2	3
3	3
4	1

VII

$y = x^2$

d more than one output number for an input number (x=1/2 has output y=+sqrt(1-1/4) or y=-sqrt(1-1/4))

$x^2 + y^2 = 1$

34. Which graph in Figure 1.13 best matches each of the following stories? Write a story for the remaining graph.

(a) I had just left home when I realized I had forgotten my books, so I went back to pick them up (IV)

(b) Things went fine until I had a flat tire. (II)

(c) I started out calmly but sped up when I realized I was going to be late. (III)

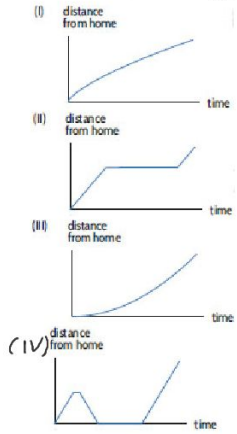


Figure 1.13

(I) I started out fast but then got tired so I got progressively slower

13. Match the graphs in Figure 1.10 with the following equations. (Note that the x and y scales may be unequal.)

(a) $y = -2.72x$ (V)
(b) $y = 0.01 + 0.001x$ (VI)
(c) $y = 27.9 - 0.1x$ (I)
(d) $y = 0.1x - 27.9$ (IV)
(e) $y = -5.7 - 200x$ (III)
(f) $y = x/3.14$ (II)

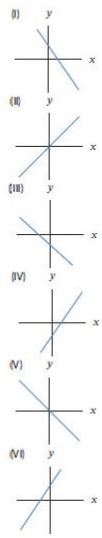
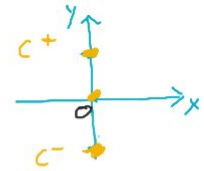


Figure 1.10

Line $y = mx + c$

Choose $x = 0 \Rightarrow y = c$

c is named y-intercept



Choose

i) $x = x_1$, $y = y_1$

$\Rightarrow y_1 = mx_1 + c$

ii) $x = x_2$, $y = y_2$

$\Rightarrow y_2 = mx_2 + c$

Balance i) and ii) by c

$c = y_2 - mx_2 = y_1 - mx_1$

$y_2 - y_1 = mx_2 - mx_1$

Name slope $:= \frac{y_2 - y_1}{x_2 - x_1} = m$

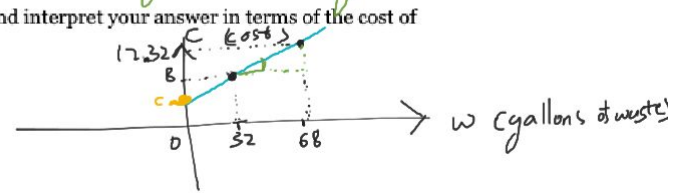
53. In a California town, the monthly charge for waste collection is \$8 for 32 gallons of waste and \$12.32 for 68 gallons of waste.

(a) Find a linear formula for the cost, C, of waste collection as a function of the number of gallons of waste, w. $C = mw + c$

(b) What is the slope of the line found in part 53(a)? Give units and interpret your answer in terms of the cost of waste collection. The rise in cost C is proportional to the change in waste by m.

(c) What is the vertical intercept of the line found in part 53(a)? Give units and interpret your answer in terms of the cost of waste collection. Base monthly charge is c

I (in)	O (out)
32	8
68	12.32



Line $y = mx + c$

Choose i) $x = x_1 = 32$, $y = y_1 = 8 \Rightarrow 8 = m\,32 + c$

ii) $x = x_2 = 68$, $y = y_2 = 12.32 \Rightarrow 12.32 = m\,68 + c$

Balance i) and ii) by c

$c = 8 - m\,32 = 12.32 - m\,68 \Rightarrow 12.32 - 8 = m\,(68 - 32)$

$\Rightarrow$ Slope $:= m = \frac{12.32 - 8}{68 - 32}$ $\left(\frac{\$}{\text{gallons}}\right)$

$\Rightarrow c = 8 - \left(\frac{12.32 - 8}{68 - 32}\right) 32$ vertical intercept (\$)

33. The value of a car, $V = f(a)$, in thousands of dollars, is a function of the age of the car, a, in years.

(a) Interpret the statement $f(5) = 6$.

(b) Sketch a possible graph of V against a. Is f an increasing or decreasing function? Explain.

(c) Explain the significance of the horizontal and vertical intercepts in terms of the value of the car.

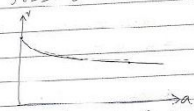
Jan board 11 5

33 $V = f(a)$ → V: value in thousands of dollars, a: age of car in years

a) If the age of the car is 5 years, its value is thou

$f(5) = 6$

b)



f is a decreasing function.

As the car gets old, its value falls.

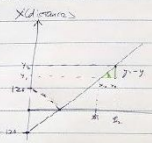
c) The horizontal intercept (x_1) is the age when the value of the car has fallen to zero. i.e. the car has no value.

The vertical intercept (y_1) is the value when the car is newly made.

45. You drive at a constant speed from Chicago to Detroit, a distance of 275 miles. About 120 miles from Chicago you pass through Kalamazoo, Michigan. Sketch a graph of your distance from Kalamazoo as a function of time.

Jan board 11 6/8

45. X(distance) constant speed: $\frac{\text{distance}}{\Delta \text{time}}$ = m (slope)

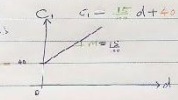


(Hint: y = X)

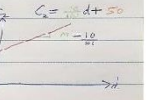
$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{X_2 - X_1}{t_2 - t_1}$

7/8

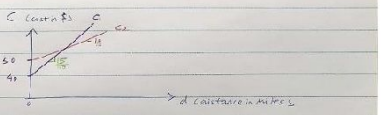
67. (a) C_1 $C_1 = \frac{15}{100} d + 40$



C_2 $C_2 = \frac{10}{100} d + 50$



(b)



(c) Balance: $\frac{15}{100} d^* + 40 = \frac{10}{100} d^* + 50 \rightarrow C^* = C_1 = C_2$

$\frac{15}{100} d^* - \frac{10}{100} d^* = 50 - 40 \Leftrightarrow \frac{5}{100} d^* = 10 \Leftrightarrow d^* = 200$ (in miles)

For $d < 200$, choose C_1 (cheaper), For $d > 200$ choose C_2

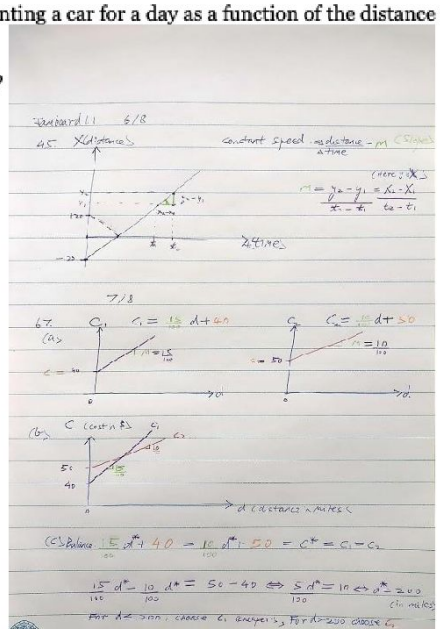
see next page 7/8 for Q68

67. A company rents cars at \$40 a day and 15 cents a mile. Its competitor's cars are \$50 a day and 10 cents a mile.

(a) For each company, give a formula for the cost of renting a car for a day as a function of the distance traveled.

(b) On the same axes, graph both functions.

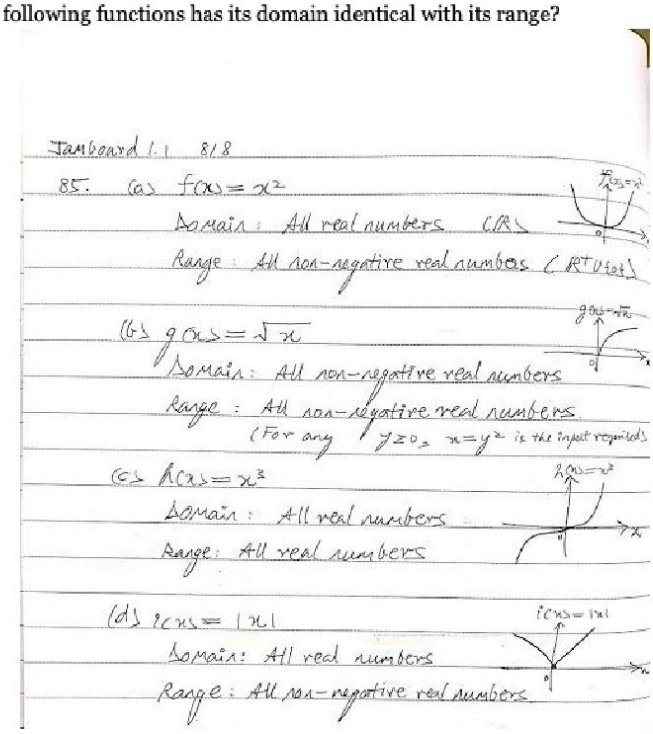
(c) How should you decide which company is cheaper?



85. Which of the following functions has its domain identical with its range?

(a) $f(x) = x^2$

(b) $g(x) = \sqrt{x}$

(c) $h(x) = x^3$

(d) $i(x) = |x|$